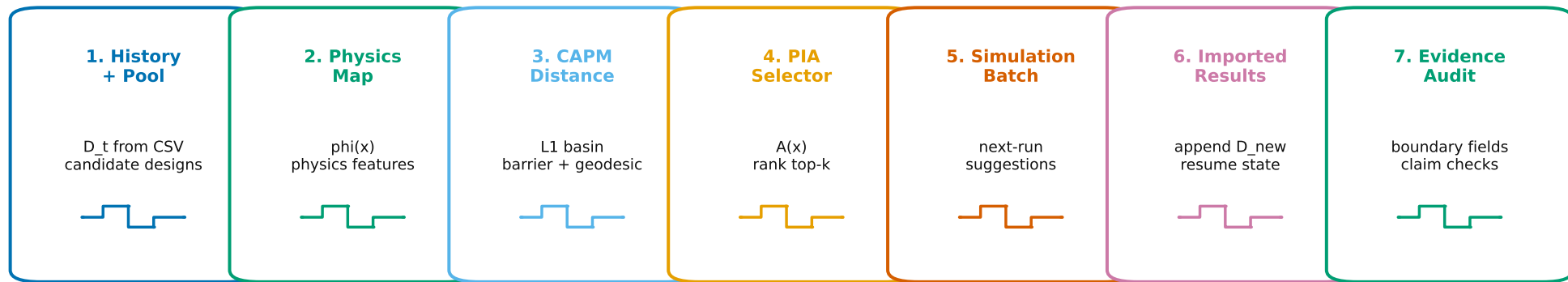


PIA-CA-LLSO Graphical Abstract

Simulation-only closed-loop GOA circuit optimization



Close the loop: append new simulation evidence and repeat until budget/target/patience stops

Boundary: data_source = real_simulation_csv | engineering_validity = simulation_only | must_resimulate = true